

Hulle_Madeline_Individual_Assignment_05

Madeline Hulle

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Set Up

Packages

```
# general use and cleaning

library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)

# coloring
library("wesanderson")
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#survey data (reading different sheets in as seperate data frames)

aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information

sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quaity

water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # values in df that are actually indicating missing values,
  # making it seem as a chr when it should be num
  # different ways missing values show up in data set
  na = c("over", "", "n/a", "N/A", "173+")
)
```

Class Code

2. Cleaning

sites object

```
# creating new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sites
```

```
filter(sector == "NCOS" &
       sampling_frequency == "Quarterly")
```

clean taxon_list

```
# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # converted taxon names to snake case
  mutate(verbatim_name = to_any_case(verbatim_name,
                                     case = "snake"))
```

clean water_quality

```
# creatign new object from water_quality
water_quality_clean <- water_quality |>
  # cleaning column name
  clean_names() |>
  # a filtering join, only keep the rows in wq clean in which the
  # site matches up with the site in ncos clean
  semi_join(ncos_sites, by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating the medians
  summarize(med_pH = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping df
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

```

# create a new object
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join only include quarterly ncos surveys
semi_join(ncos_sites, by = "site") |>
# renaming date_on_vial column to date
rename(date = date_on_vial) |>
# selecting columns of interest
# keeping information about each survey and the 9 most abundant taxa
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filtered to only include filteres beaker samples
# (better sampling technique)
filter(sample_type %in% c("FB 250", "FB250")) |>
# converting to long format, making df longer
pivot_longer(
  # name the columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for old column names
  # column names are all in new column called taxon
  names_to = "taxon",
  # new column for the values in each of the old columns
  # counts now under a column called abundance
  values_to = "abundance"
) |>
# group by date site and taxon
group_by(date, site, taxon) |>
# calculated average count per liter (total abundance/7.7)
summarise(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungrouping data frame
ungroup() |>
# new column to join data frames, didnt remove them
unite("date_site", date, site, remove = FALSE) |>

```

```

# joined with water quality clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# set factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
       # reversing the order
       site_full = fct_rev(site_full))

```

Visualizing

Salinity

```

# creating new object to store plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
# points representing median salinity at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points representing mean salinity
stat_summary(geom = "point",
            fun = median,
            color = "red",
            size = 3) +

```

```
# adding line breaks to x axis labels (wrapping)
scale_x_discrete(labels = label_wrap(width = 15)) +
# relabeling axes and adding a title
labs(x = "Site",
      y = "Median Salinity (ppt)",
      title = "Site Salinity") +
# adding themeing
theme_bw()
```

Copepod Abundance

Cleaning

```
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  # adding 1 bc some observations are 0
  # helps overcome issues with large differences in order of magnitude
  mutate(log_ave_lit = log(ave_lit + 1))
```

Copepod Viz

```
# creating object to store plot
copepod_plot <- ggplot(data = copepods,
                       mapping = aes(x = site_full,
                                      # using log transformed count per liter
                                      y = log_ave_lit)) +

# jitter to represent each sample
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# points to represent median copepod abundance
stat_summary(geom = "point",
             fun = median,
             color = "blue",
             size = 3) +
# wrappign x-axis labels
scale_x_discrete(labels = label_wrap(width = 10)) +
# relabeling axes
labs(x = "Site",
      y = "Log +1 transformed average abundance/L ",
```

```

    title = "Copepod Abundance") +
# using different theme
theme_bw()

```

Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: `site_full`
- y-axis: `log_ave_lit` (updated w/ `log_ave_lit` after viz)
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary(geom = "point", fun = median)`

b. Wrangle your data

```

# creating a new object `ostracods` from `aq_ins_clean`
ostracods <- aq_ins_clean |>
  # filtering to only include ostracods
  filter(taxon == "ostracod") |>
  # log transforming `ave_lit` to overcome large difference in
  # order of magnitude between observations
  mutate(log_ave_lit = log(ave_lit + 1))

# taking a sample of 10 observations
slice_sample(ostracods, n = 10)

```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_pH	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.267	NA	NA	NA
2	2023-06-06_NPB1	2023-06-06 00:00:00	NPB1	ostr~	0.533	7.48	NA	2.38
3	2022-08-12_NEC	2022-08-12 00:00:00	NEC	ostr~	0	7.37	NA	NA
4	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	ostr~	0	8.86	45.2	10.2
5	2023-02-21_NEC	2023-02-21 00:00:00	NEC	ostr~	0	8.56	NA	8.8
6	2023-01-28_NPB1	2023-01-28 00:00:00	NPB1	ostr~	0	7.66	NA	9.74
7	2022-02-25_NEC	2022-02-25 00:00:00	NEC	ostr~	0	8.75	44.1	9.87

```

8 2021-08-25_NPB 2021-08-25 00:00:00 NPB ostr~ 581.      7.52    4.57    4.5
9 2023-11-14_NPB 2023-11-14 00:00:00 NPB ostr~ 18        NA      NA      NA
10 2023-05-16_NVBR 2023-05-16 00:00:00 NVBR ostr~ 0.133    9.29    NA      9.83
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

c. Code Your Figure

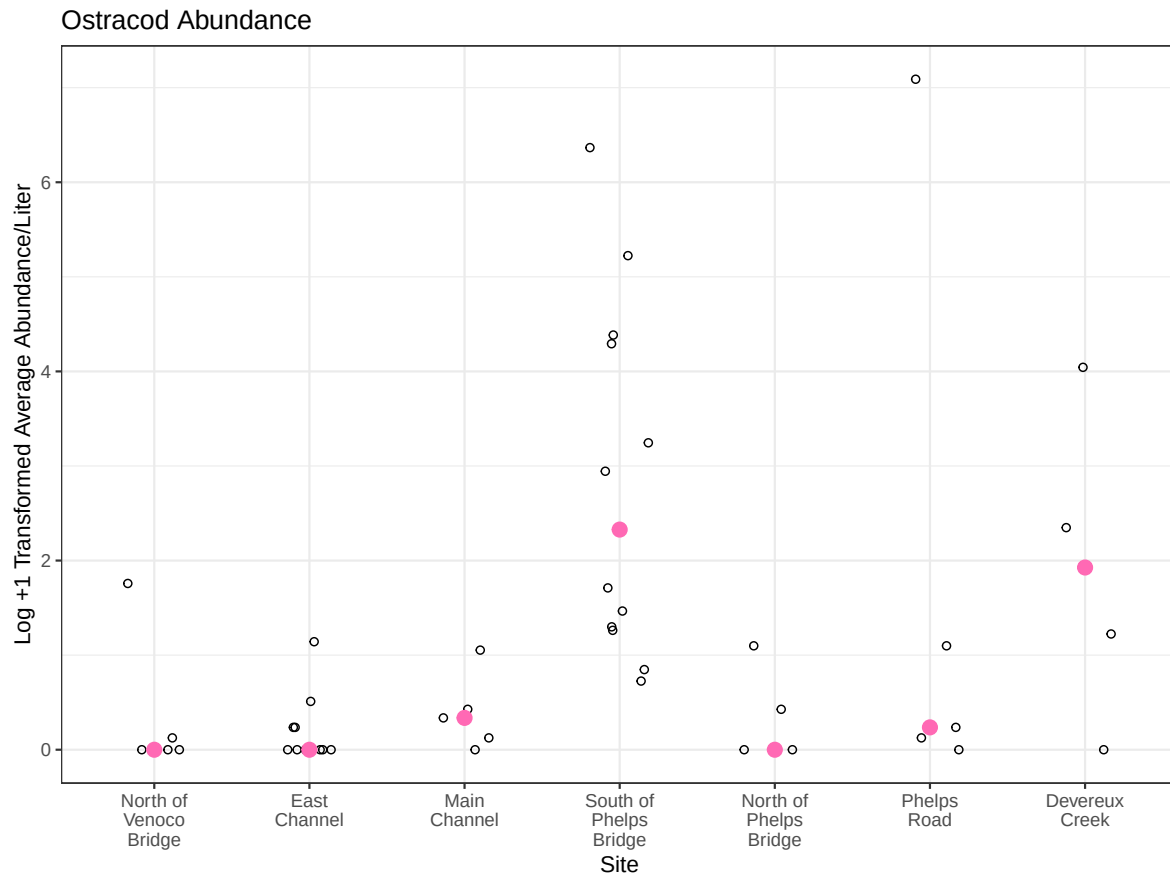
```

ostracod_plot <- ggplot(data = ostracods,
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +

  geom_jitter(height = 0,
              width = 0.2,
              shape = 21) +
  stat_summary(geom = "point",
              fun = median,
              color = "hotpink",
              size = 3) +
  labs(x = "Site",
       y = "Log +1 Transformed Average Abundance/Liter",
       title = "Ostracod Abundance") +
  theme_bw() +
  scale_x_discrete(labels = label_wrap(width = 10))

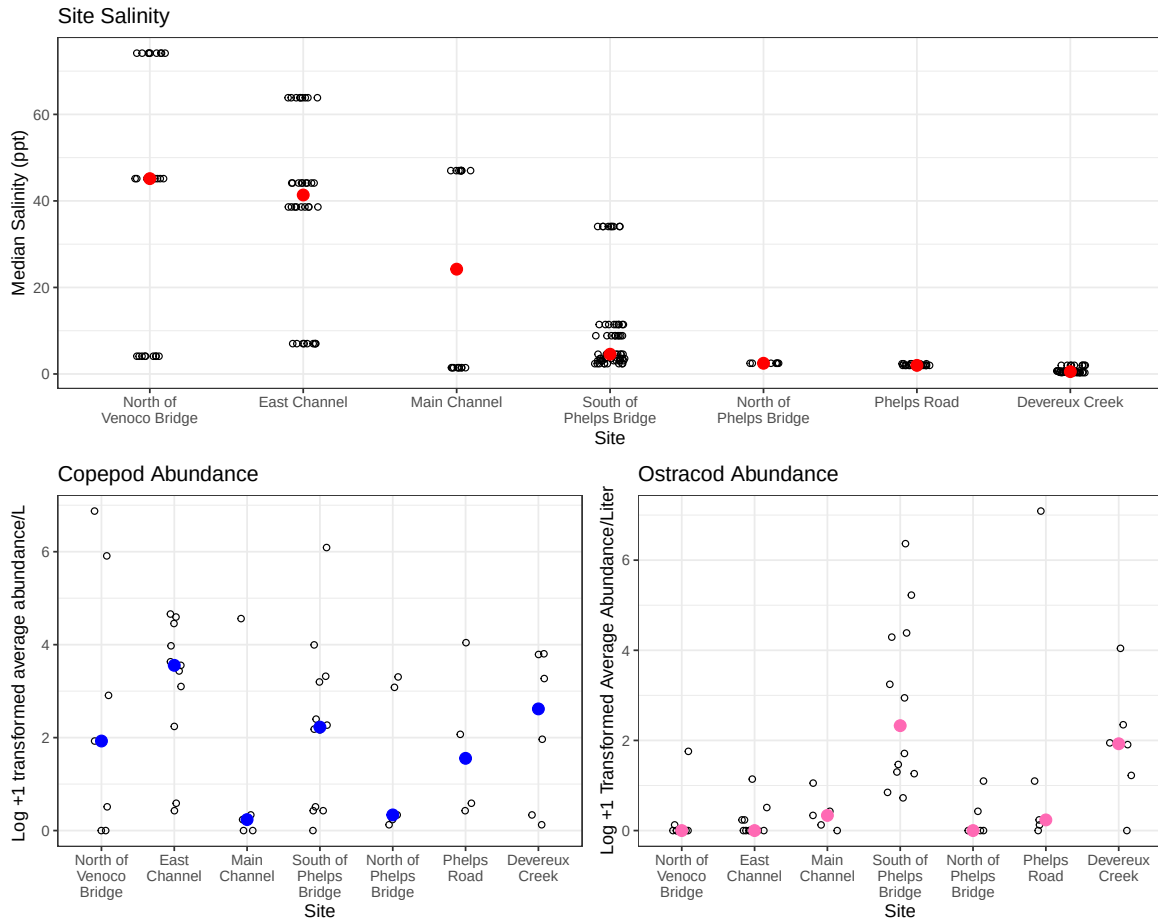
ostracod_plot

```

d. Create a Multi Panel Figure

```
salinity_plot / (copepod_plot | ostracod_plot)
```



e. Interpret Your Figure

- Between sites there does seem to be a difference between the median log + 1 average abundance/L of Copepods (blue dots) and that of Ostracods (pink dots). For sites North of Venoco Bridge and East Bridge, the median log + 1 average abundance/L of Ostracods is zero while for Copepods, this value is greater than zero, with East Channel being the site with the greatest median log + 1 average abundance/L of all of the sites. For the Main Channel site, Ostracod and Copepod median log + 1 average abundance/L are quite similar, both slightly above 0. The site South of Phelps Bridge also sees similar levels of median log + 1 average abundance/L across both groups, with this being the highest median log + 1 average abundance/L seen for the Ostracods. At North of Phelps Bridge we see low median log + 1 average abundance/L for both groups, with Ostracod having a slightly lower median log + 1 average abundance/L than Copepods. Finally, Phelps Road and Devereux Creek see similar relative median log + 1 average abundance/L values

across the two groups, but it is higher for Copepods at both sites. Overall, there seems to be, on average, higher Copepod abundance across these 7 sites than Ostracod abundance as more of the blue (median log + 1 average abundance/L for Copepods) points are higher up on the y-axis than the pink ones (median log + 1 average abundance/L for Ostracods).

- Comparing the Site Salinity plot to the Ostracod Abundance plot, there seems to be a possible relationship between the two. For site salinity, the mean salinity level is highest for North of Venoco Bridge and steadily decreases through the intermediate sites until the lowest median salinity is recorded at Devereux Creek. In the Ostracod Abundance plot, we see an almost inverse pattern where North of Venoco Bridge and East Channel see the lower median log + 1 average abundance/L levels and this steadily increases (with the exception of the South of Phelps Bridge observation) along the intermediate sites and reaching a peak at Devereux creek. More data would need to be collected and more analysis be done to determine if this is a true relationship and if the South of Phelps Bridge observation is a fluke) or if these inverse shapes are a coincidence in the data. From my visualization now however, it seems possible that higher salinity corresponds with a lower abundance of Ostracods and a lower salinity corresponds to a higher abundance of Ostracods. The optimal environmental conditions of Ostracods would also be helpful information to determine the validity of this relationship.

2. Visualizing total abundance as a function of salinity

a. Plan Your Figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_jitter()
- facets: facet_wrap(~taxon)

b. Wrangle Your Data

```
# creating a new object `log_aq_ins_clean` from `aq_ins_clean`
log_aq_ins_clean <- aq_ins_clean |>
  # log transforming the `ave_lit` column
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # turnign each taxon into title case
  mutate(taxon = to_any_case(taxon,
                             case = "title")) |>
  # reordering the levels of taxon by the max abundance
```

```
mutate(taxon = fct_rev(fct_reorder(taxon,
  ave_lit,
  .fun = max,
  na.rm = TRUE)))
```

```
# taking a sample of 10 observations
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_pH <dbl>	med_sal <dbl>	med_do <dbl>
1	2022-05-06_NMC	2022-05-06 00:00:00	NMC	ostr~	0	9.96	1.45	16.5
2	2022-02-23_NVBR	2022-02-23 00:00:00	NVBR	ostr~	0	8.86	45.2	10.2
3	2022-11-18_NVBR	2022-11-18 00:00:00	NVBR	ostr~	0	NA	NA	NA
4	2022-05-11_NDC	2022-05-11 00:00:00	NDC	ostr~	0	9.11	0.398	9.4
5	2022-02-23_NPB1	2022-02-23 00:00:00	NPB1	ostr~	0	NA	NA	NA
6	2022-05-12_NPB	2022-05-12 00:00:00	NPB	ostr~	1.07	7.97	11.4	8.17
7	2021-11-16_NPB	2021-11-16 00:00:00	NPB	ostr~	72.1	7.3	2.38	12.1
8	2022-11-12_NEC	2022-11-12 00:00:00	NEC	ostr~	0	7.33	7.01	2.4
9	2022-05-12_NPB2	2022-05-12 00:00:00	NPB2	ostr~	1198	NA	NA	NA
10	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.2	NA	NA	NA

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code Your Figure

```
# creating a new object to store plot from the data `log_aq_ins_clean`
log_abundance_salinity <- ggplot(data = log_aq_ins_clean,
  # x-axis: median salinity
  # y-axis: log transformed average abundance/L
  mapping = aes(x = med_sal,
    y = log_ave_lit,
    # coloring by taxon
    color = taxon)) +

# points representing median salinity
geom_jitter(height = 0) +
# plot for each taxon with own y-axis
facet_wrap(~ taxon,
```

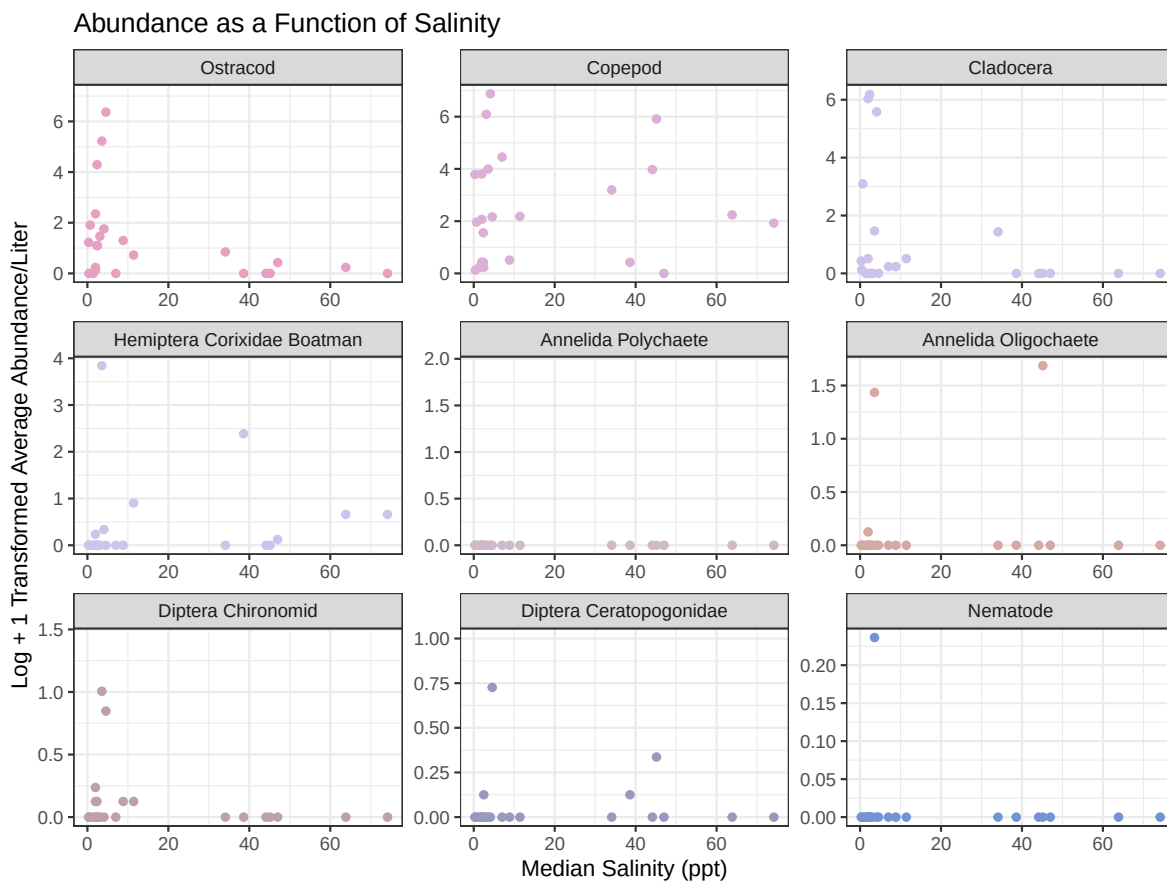
```

    scales = "free") +
# adding in custom coloring
scale_color_manual(values = wes_palette("GrandBudapest2",
                                         n = 9, type = "continuous")) +

# custom theme
theme_bw() +
# taking out legend, redundant
theme(legend.position = "none") +
# labeling x and y axes
labs(x = "Median Salinity (ppt)",
     y = "Log + 1 Transformed Average Abundance/Liter",
     # adding a title
     title = "Abundance as a Function of Salinity")

# displaying the plot
log_abundance_salinity

```



d. Interpret Your Figure

Ostracod, Copepod, and Cladocera show the highest log-transformed abundances overall (y-axes reaching 6–7), and all three display a clear negative relationship with salinity as observations cluster heavily at low salinity (0–10 ppt) with higher abundance values, a pattern that decreases with increased salinity values. Hemiptera Corixidae Boatman follows a similar pattern of low-salinity concentration but at much lower maximum abundances (~4). The annelid taxa (Polychaete and Oligochaete) and the dipteran taxa (Chironomid and Ceratopogonidae) are present across the full salinity range (0–70 ppt), suggesting stronger salinity tolerance, though their abundances remain consistently low regardless of salinity. Nematode stands out as the taxon with the lowest maximum abundance of all nine (~0.25), with observations scattered across salinity levels and no discernible trend. Overall, most taxa appear to prefer low-salinity conditions, but the top three taxa, Ostracod, Copepod, Cladocera, show this preference most dramatically in both abundance magnitude and distributional pattern.